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Hydraulic Fracture Propagation Along Bedding Planes Might be More Prevalent than We Think

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Abstract

It is commonly assumed that most commercial unconventional plays exhibit predominantly vertically oriented hydraulic fractures. At the same time, there is mounting evidence that hydraulic fractures may propagate along sub-horizontal bedding planes in some instances. This paper first summarizes field observations of horizontal hydraulic fractures, which include measurements from tiltmeters, microseismic, fiber optics, casing deformation, as well as observations of propped horizontal fractures in a vertical core. Then, rock properties are revisited with respect to their influence on horizontal fractures. Shales are finely layered rocks that typically exhibit elastic, toughness, and tensile strength anisotropies. Tensile strength anisotropy in particular makes it easier to initiate a sub-horizontal fracture along a bedding interface, rather than a vertical fracture perpendicular to layers. As a result, it is possible to initiate a horizontal hydraulic fracture even if vertical stress exceeds its horizontal counterpart. This observation is supported by laboratory experiments. In addition to this, shales exhibit viscoelasticity or creep, which leads to stress relaxation and brings the state of stress close to hydrostatic. Note that the latter nearly hydrostatic stress state can occur in certain layers regardless of the stress regime. The combination of viscous stress relaxation and tensile strength anisotropy greatly elevates the possibility of initiation of horizontal hydraulic fractures. Operational parameters may also contribute to the creation of horizontal fractures. As single well stimulations are being replaced with multi-well developments, this leads to an overall higher stress build-up in the reservoir. In particular, higher fluid loadings and closer well spacings lead to bigger stress change. As the stress shadow accumulates and the minimum horizontal stress increases, at some point the stress contrast becomes insufficient and horizontal fractures start to form. We provide estimates to evaluate operational parameters that influence the stress build up in the formation. It is shown that horizontal hydraulic fractures are affected by both rock and operations. The rock has a certain capacity or stimulation intensity limit before horizontal fractures start to form. Therefore, operations need to be tailored to a specific location, if the goal is to avoid horizontal fractures. Several examples illustrating formation of horizontal fractures are presented.

Introduction

Hydraulic fracturing is a technique that is widely used to stimulate oil and gas reservoirs (Economides and Nolte, 2001; King, 2010). Starting from the early days, fracture geometry has been a key topic for debates. It is widely recognized that hydraulic fractures propagate perpendicular to the minimum principal stress in the subsurface (Hubbert and Willis, 1957). For sufficiently deep reservoirs, the minimum principal stress is horizontal and therefore hydraulic fractures are predominantly vertical. There is a very strong body of field evidence supporting this. For instance, recent fiber optic measurements clearly indicate the presence of vertical and mostly planar hydraulic fractures (Shahri et al., 2022; Ugueto et al., 2021 and 2023). Nevertheless, there are a variety of scenarios where evidence suggests that horizontal fractures form and propagate. This paper aims to address this topic with the focus on unconventional rocks.

It should be noted that hydraulic fractures are also used in non-petroleum applications. For instance, the use of hydraulic fractures has been proposed for energy storage (Bunger et al., 2023). Because the fluid is injected at a relatively shallow depth for such applications, the induced fractures are nearly horizontal, which is consistent with the fact that vertical stress is the least principal stress at shallow depths. Similarly, horizontal fractures are observed in mining applications, in which fracturing is used for rock preconditioning (Jeffrey and Mills, 2000). Again, shallow depth and the smallness of the vertical stress explain this phenomenon. These examples demonstrate that it is possible to have large scale horizontal hydraulic fractures in rocks, i.e. hydraulic fractures underground are not always vertical.

One of the pioneering papers that discussed the presence of horizontal fractures for petroleum applications dates back almost three decades ago (Wright et al., 1997). The authors provide tiltmeter data that indicates potential formation of horizontal hydraulic fractures, or at least horizontal fracture component, during conventional waterflooding operations. In particular, a case study from the North Shafter Field (USA) suggests the formation of horizontal fractures at 7500 ft depth. Not all wells exhibited the horizontal fracture component, but rather a non-negligible proportion. Another example in the same paper is presented for the Belridge Oil Field (USA), in which enhanced Oil Recovery (EOR) is believed to change the state of stress, which led to the formation of horizontal fractures. Arguably, tiltmeters can be affected by the thermoporoelastic deformations caused by the long term injection, but, nevertheless, these examples indicate that horizontal fractures can also be influenced by operations, such as long term injection or EOR. This paper demonstrates that horizontal fractures are caused by either abnormally low vertical stress, or, more likely, by the stress change caused by operations.

More recently, there have been published examples of horizontal fracturing in shale formations evidenced by microseismic and treating pressures, core evaluation, and fiber optic strain. Alalli and Zoback (2018) report formation of horizontal hydraulic fractures in Marcellus Shale (USA) using microseismic measurements, see Fig 1. In particular, results indicate that the microseismic cloud is very height contained at the toe of the well, which is interpreted as the formation of horizontal hydraulic fractures. Interestingly, the microseismic events show noticeable height growth towards the heel of the well. This is explained by the fact that the well meanders to a different rock formation at the heel and this rock has a smaller value of the horizontal stress. Such a phenomenon was reported for several nearby wells, which confirms that this is not an artifact of the measuring system. In addition, Instantaneous Shut-In Pressure (ISIP) values were above the estimated vertical stress for the toe portion of the well that corresponds to height contained microseismic events or horizontal fractures. At the same time, the ISIP values were below the vertical stress for the heel part of the well, which is consistent with the formation of vertical hydraulic fractures. This example indicates that neighboring rock formations can potentially lead to very different fracture geometries, and thus rock matters. Also, high values of ISIP can be indicative of formation of horizontal hydraulic fractures.

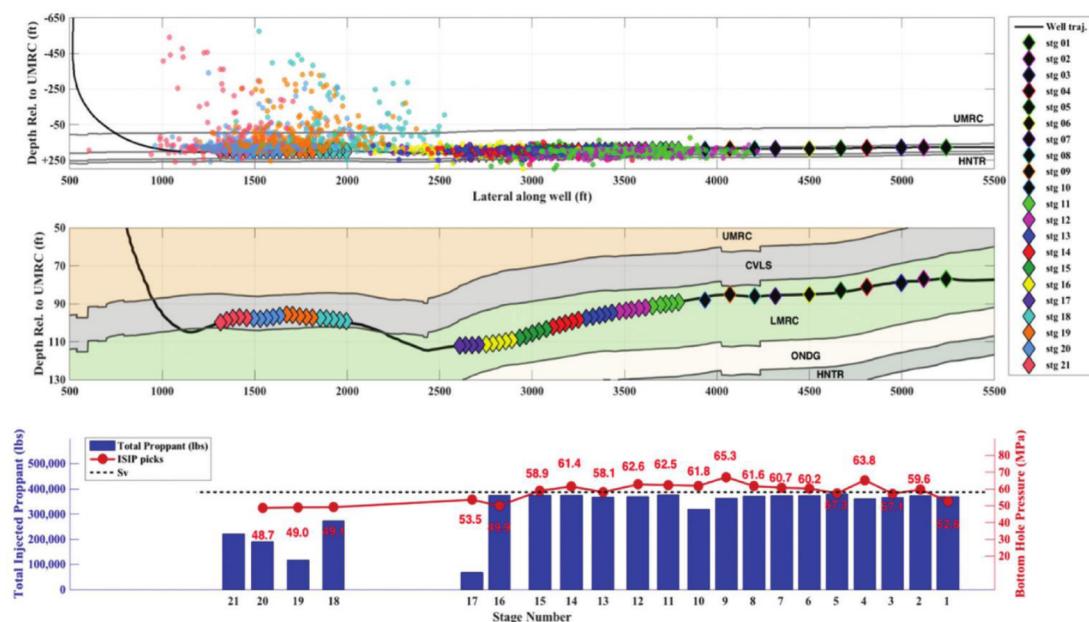


Figure 1—Observation of horizontal hydraulic fractures in Marcellus shale. Top: profile view of microseismic events. Middle: well trajectory. Bottom: injected proppant (blue bars), ISIP (red circular markers), and vertical stress (dashed black line). Reproduced from Alalli and Zoback (2018) in accordance with publisher guidelines.

There is yet further evidence of the presence of horizontal hydraulic fractures in shales, see Fig. 2. A vertical well was drilled and cored outside of the edge of the microseismic cloud at a pad in Eagle Ford (USA). The microseismic cloud indicated substantial vertical height growth throughout the whole pad, while the vertical well, again, was drilled slightly outside of the main microseismic cloud. Surprisingly, a propped horizontal fracture was observed in the core. Since hydraulic fracture length typically noticeably exceeds propped length, then we can anticipate that lateral extent of the horizontal fracture was significant in this pad. Microseismic typically shows fracture half-lengths of 500-1000 ft. Since the propped horizontal fracture was observed slightly further, then it is natural to expect that the hydraulic horizontal fracture size was several 1000s ft. This observation shows that, first of all, horizontal fracture growth was invisible to microseismic in this case, which is in contrast to the previous Marcellus example (Alalli and Zoback, 2018). One possible explanation is that horizontal fractures open up along weak interfaces, which do not produce strong signals during delamination. Another striking observation is that Eagle Ford has a normal faulting stress regime, whereby the minimum horizontal stress is the lowest, while the vertical stress is the highest (Snee and Zoback, 2022). This means that horizontal hydraulic fractures opened against the initial, in-situ direction of the maximum stress! A detailed explanation for this example will be presented later in the paper. This case demonstrates that both vertical and horizontal fractures can coexist within the same pad and that horizontal fractures can extend large distances, open up, and transport proppant.

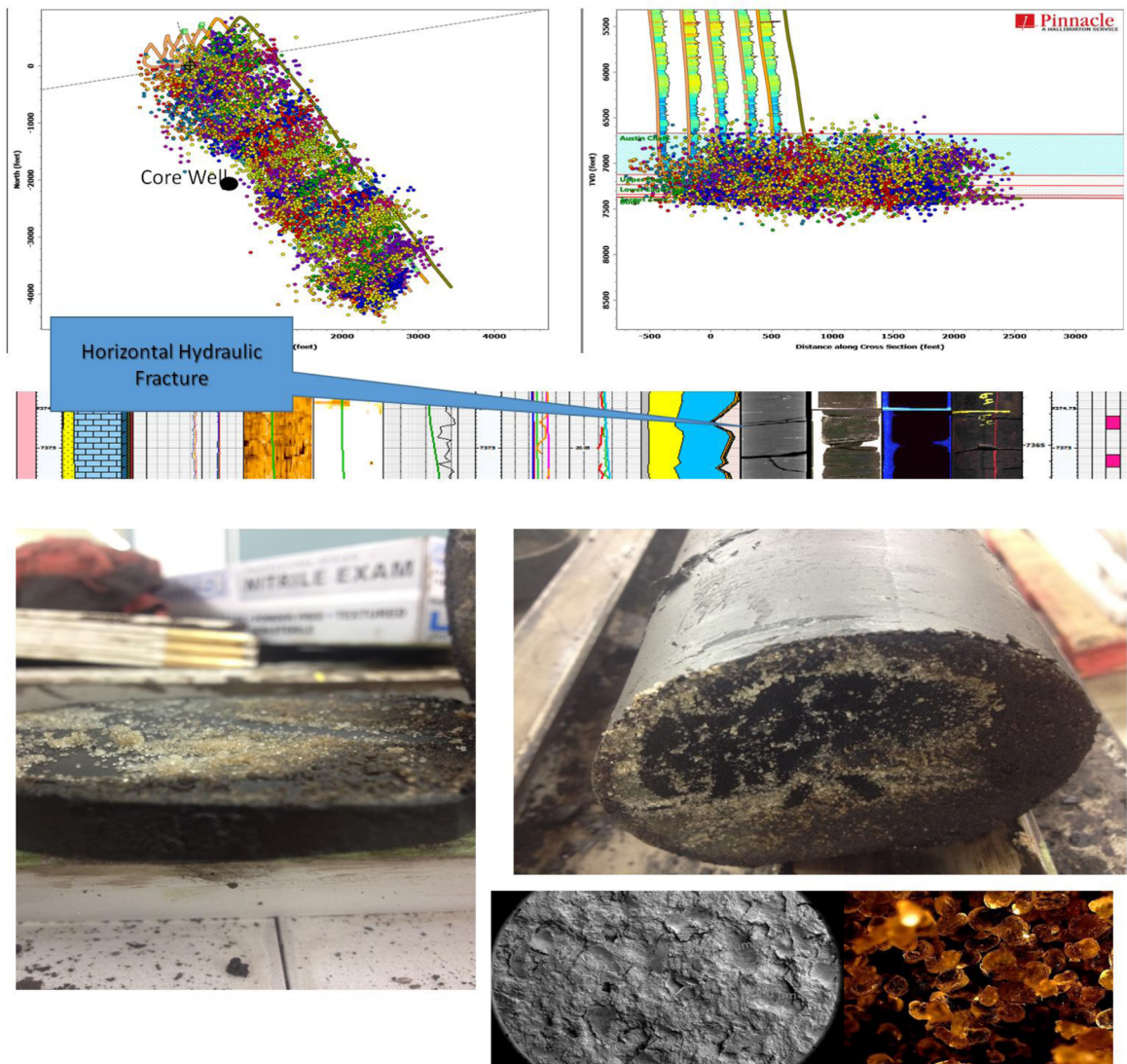


Figure 2—Observation of a propped horizontal hydraulic fracture in Eagle Ford. Top: microseismic events from stimulation and location of the cored vertical well. Bottom: photographs of the propped horizontal fracture found in the core. Courtesy of Von Gonten Engineering.

A very recent paper (Wang et al., 2025) presented fiber optic measurements of sub-horizontal hydraulic fractures in Sichuan Basin (China), see Fig. 3. Authors reported fiber optic signals observed in an offset horizontal well. Results indicate that there were several vertical fractures and a sub-horizontal fracture overlaid on top of them. Modeling of fiber optic signals with a boundary element approach indicates that the sub-horizontal fracture had both opening and sliding components. Such behavior was persistent for several stages and at some point the sub-horizontal fracture crossed the well. Interestingly, similarly to the previous example in Eagle Ford, both vertical and horizontal fractures propagate simultaneously or coexist together. Also, the data provides evidence that horizontal fractures have an opening, which is consistent with the propped horizontal fracture discussed in the previous example.

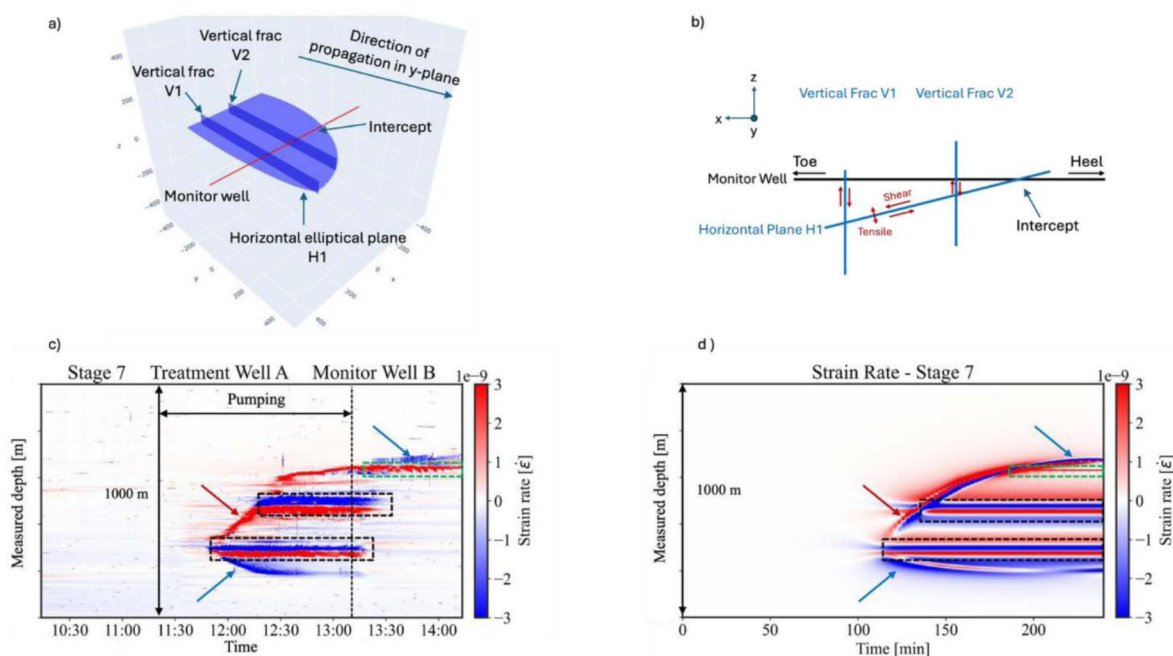


Figure 3—Observation of a horizontal hydraulic fracture in Sichuan Basin. (a): 3D schematic of observations, red line shows the monitor well, while the blue regions indicate vertical and horizontal fractures. (b): Plan view of observations, fracture opening and sliding directions are shown. (c): Signal observed in the field. (d): Simulated signal, assuming opening and sliding of the horizontal fracture. Reproduced from Wang et al. (2025) with publisher's permission.

While the hydraulic fracture geometry has a fundamental interest for many researchers, it also bears strong practical implications. First of all, the vertical permeability of shales is known to be much lower than its horizontal counterpart due to fine laminations. As a result, surface area created by horizontal fractures will contribute less to production. Once a horizontal fracture forms, there is potential to create less vertical height growth as other vertical hydraulic fractures can be arrested by an opened horizontal fracture. Proppant transport in horizontal fractures is likely very different from that in vertical fractures because particles will essentially roll over the horizontal fracture surface, so it will be dune-like transport. Since horizontal fractures can span large distances and do not have natural barriers along their way (e.g. high stress layers), they can cause hydraulic connections between different stages and/or different wells. Finally, since horizontal fractures often exhibit slip deformation, they can cause casing deformation. Note that the slip due to pressurization of an existing mechanically closed discontinuity is limited by the fluid diffusion along the discontinuity and is therefore spatially constrained. At the same time, an open horizontal fracture has very high conductivity, propagates large distances, and is unable to sustain shear stress. Therefore, an open horizontal fracture has a much higher potential to cause severe slip and casing deformation. There is plenty of field evidence of casing deformation caused by bedding plane slippage in different geologic locations, see e.g. a review paper Uribe-Patino et al. (2024). A horizontal fracture has potential to deform the vertical part of the well, which can lead to significant consequences for field operations.

To summarize field observations, while hydraulic fractures in shales are often considered to be vertical and there is evidence for this, there is also a mounting body of evidence demonstrating the presence of horizontal hydraulic fractures. A combination of vertical and horizontal fractures is observed in multiple shale formations. Horizontal fractures open up, are able to carry proppant, and also cause slip as they are unable to sustain shear stress. They do not have natural barriers and can grow large in size and by doing so they ‘transport’ slip and can cause casing deformation at large distances away from the injection point. Horizontal fractures are clearly affected by the rock type as well as operations since the latter can noticeably change the state of stress. While most of the field evidence points to formations with a nearly hydrostatic state of stress and strike-slip stress regime, it is possible to have horizontal fractures in formations that

initially are in normal faulting. The rest of the paper aims to investigate this phenomenon and to explain some of the behavior observed in the field. As was mentioned above, horizontal fractures are influenced by both rock and operations. Both of these aspects are covered next.

The influence of rock on horizontal fractures

For the purpose of hydraulic fracture modeling, rock is often considered to be isotropic and linear elastic. This is a suspect assumption in shales, which are highly laminated on multiple scales (Suarez-Rivera et al., 2016). The small scale layers introduce elastic anisotropy, whereby Young's modulus in the horizontal direction is different from that in the vertical direction. There is also fracture toughness anisotropy, which leads to a stronger resistance to vertical fracture growth across the layers relative to the lateral fracture growth along the layers. Both of these phenomena were investigated in the past in the context of vertical hydraulic fractures (Bessmertnykh and Dontsov, 2018; Lu et al., 2024). Results highlight weak dependence of vertical fractures on elastic anisotropy, while noticeable dependence of fracture geometry on fracture toughness contrast. In particular, for relatively low viscosities, the toughness anisotropy leads to a more elongated fracture shape with shorter height and longer length.

If a horizontal fracture propagates in such anisotropic formation, then its net pressure is affected by the vertical Young's modulus. To confirm that this statement is true, Fig. 4 plots the effective elastic modulus for a horizontal fracture in a transversely isotropic material (Laubie and Ulm, 2014) versus that for an isotropic material defined by E_v and ν_v , i.e. vertical Young's modulus and vertical Poisson's ratio. Expression for the effective modulus in a transversely isotropic material is given by (Laubie and Ulm, 2014)

$$H_1 = \frac{1}{2\pi} \sqrt{\frac{C_{11}}{C_{11}C_{33}-C_{13}^2} \left(\frac{1}{C_{44}} + \frac{2}{C_{13} + \sqrt{C_{11}C_{33}}} \right)}, \quad E'_{aniso} = \frac{1}{\pi H_1},$$

where C_{ij} are elastic constants. At the same time, the corresponding modulus for an isotropic material is

$$E'_{iso} = \frac{E_v}{1 - \nu_v^2}.$$

Fig. 4 plots data (square markers) measured for anisotropic rock samples, reported in literature (Chertov, 2012; Laubie and Ulm, 2014a; Jin et al., 2018; Johnston and Christensen, 1994; Sondergeld and Rai, 2011). The black solid line corresponds to $E'_{aniso} = E'_{iso}$. As can be seen from the results, horizontal hydraulic fractures are indeed predominantly affected by the vertical Young's modulus, even though there are some minor deviations. The latter deviations come from the fact that the deformation pattern around a horizontal fracture is not strictly uniaxial.

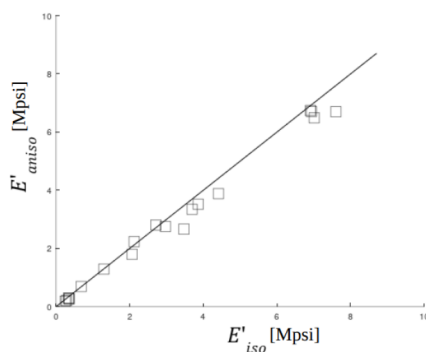


Figure 4—Effective anisotropic elastic modulus for a horizontal fracture versus its approximation based on vertical Young's modulus.

Because the vertical Young's modulus is often lower than its horizontal counterpart for shales (up to 4x in the data mentioned above), this implies that horizontal fractures have smaller net pressure, are wider and shorter relative to vertical fractures with the same volume. For instance, for a viscosity dominated radial fracture with a given volume and $E_h/E_v = 2$, the horizontal fracture is 1.17 times wider, net pressure is 1.59 times lower, while the radius is 1.36 shorter than that for a vertical fracture (Dontsov, 2016).

Resistance to horizontal fracture propagation is controlled by delamination. The associated fracture toughness is hard to measure and its value is expected to be much lower than that for a vertical fracture. The reduced toughness value leads to narrower fracture, lower net pressure, and longer size. This leads to potentially bigger horizontal fractures.

There is one more type of anisotropy that affects fracture initiation, which is tensile strength anisotropy. Hydraulic fractures initiate when fluid pressure exceeds stress plus tensile strength in a particular direction. For isotropic rocks, this immediately implies that hydraulic fractures initiate in the plane that is perpendicular to the minimum principal stress. Due to the layered nature of shales, however, the tensile strength may be greater for the initiation of vertical fractures. As a result, hydraulic fractures can initiate along a weak direction even though this is not the most favorable direction from the stress perspective. This has been observed in several published laboratory studies. For instance, Abdelaziz and Grasselli (2024) present results for a hydraulically fractured shale sample and reported horizontal fracture initiation, despite the vertical stress exceeding the minimum horizontal stress. A study by Suarez-Rivera et al. (2013) attempted to measure fracture toughness across layers, but instead the sample fractured along the bedding planes. These laboratory observations demonstrate the effect of 'rock fabric' on hydraulic fracturing, which is essentially characterized by zones in the rock with lower fracture toughness and lower tensile strength. For the purpose of horizontal hydraulic fracture modeling, the horizontal fracture initiation criterion is

$$P_w > S_v + T_v,$$

where P_w is fluid pressure, S_v is the total vertical stress, and T_v is the effective tensile strength in the vertical direction. For completeness, vertical fracture initiation criterion is

$$P_w > S_h + T_h,$$

where S_h is the total minimum horizontal stress, while T_h is the effective tensile strength in the horizontal direction. These equations demonstrate the possibility of horizontal fracture initiation against the vertical stress S_v even when $S_v > S_h$, i.e. when the vertical stress exceeds the minimum horizontal stress. This contradicts the common belief that fractures always initiate against the least principal stress. For instance, consider a situation, in which the vertical stress is $S_v = 10000$ psi and the minimum horizontal stress is $S_h = 9500$ psi. Then, if the tensile strength in the horizontal direction is $T_h = 1000$ psi, while its vertical counterpart is $T_v = 0$ psi, then $S_v + T_v = 10000 < 10500 = S_h + T_h$ and horizontal fractures will initiate.

Another mechanism that promotes horizontal hydraulic growth is viscoelastic stress relaxation (VSR). This refers to time-dependent rock deformation first observed in laboratory studies of clay- and organic-rich rocks (Sone and Zoback 2013 a,b). It accurately explains observations of lithology-dependent variations of the least horizontal principal stresses, S_h , with depth (Sone and Zoback, 2014; Kohli and Zoback 2021; Singh and Zoback, 2020; McCormack et al. 2021)). As a clay-rich rock deforms over time (viscous deformation), the differences between the three principal stresses become smaller. Even in normal faulting areas where S_h is normally much less than the vertical principal stress, the minimum horizontal stress S_h can increase so much due to VSR that it is approximately the same as the vertical principal stress, resulting in a nearly isotropic stress field. An example of such isotropic stress conditions in clay-rich formations in a normal faulting area is shown in Figure 3.19 of Zoback and Kohli (2019). In this case, numerous minifrac in three vertical wells show expected values of S_h in sands but $S_h \approx S_v$ in shales and mudstones. Zoback and Kohli (2019) discuss the viscoplastic behavior of unconventional reservoir rocks at length as well as its affect on the magnitude of S_h . In a case study similar to that shown in Fig. 1, Ma and Zoback (2017)

show lithologically-controlled variations of the least principal stress along two horizontal wells drilled in the Woodford formation in Oklahoma. The well paths (relative to < 10m thick lithofacies) resulted in some stages being stimulated under conditions of essentially isotropic stress in clay-rich lithofacies. To summarize, the way in which isotropic principal stresses contribute to the tendency for horizontal hydraulic fracture propagation is straightforward. As the stress contrast is not sufficient to control hydraulic fracture propagation, it is easiest for hydraulic fractures to propagate along bedding planes with low tensile strength. It should be noted that some of the hydraulic fracturing stages can end up in rocks with no stress anisotropy regardless of the stress regime (normal faulting, strike-slip, or reverse faulting).

In contrast to typical assumptions made in modeling, unconventional rocks are strongly anisotropic and viscoelastic. Tensile strength anisotropy promotes initiation of horizontal fractures, even in situations for which vertical stress exceeds the horizontal components. Elastic anisotropy increases compliance of horizontal fractures, i.e. less pressure is required to open a similar fracture. Toughness anisotropy reduces resistance to propagation of horizontal fractures, causing bigger fracture dimensions. Finally, viscoelasticity of rocks reduces the stress contrast in the formation, which further promotes horizontal fracture initiation and propagation. This highlights the importance of rock to horizontal fractures. Note that while some of these rock parameters are well studied, some others are investigated to a lesser extent, which presents opportunities for laboratory studies.

How operations promote horizontal fractures

Fluid injection into the subsurface inevitably leads to a change in the state of stress. Whether the fluid filtrates into the formation and increases the pore pressure or induces a fracture, both situations lead to stress increase. Conversely, fluid withdrawal or production, and rock shrinkage due to thermal effects caused by long term fluid injection (e.g. waterflooding) or circulation (e.g. geothermal applications) will reduce stress. The state of stress strongly affects initiation of horizontal fractures. This section, therefore, attempts to estimate how hydraulic fracturing operations affect the state of stress.

As a starting point, let's consider the problem of a single vertical hydraulic fracture with the given height H . According to the classical Perkins-Kern-Nordgren (PKN) model (Perkins and Kern, 1961), the fracture net pressure is

$$P_n = \frac{2EV}{HA},$$

where $E' = E/(1 - \nu^2)$ is the plane strain Young's modulus, V is the fracture volume, H is the fracture height, while A is the fracture area. Note that this equation assumes elastically isotropic formation. For the elastically anisotropic formations, horizontal Young's modulus provides a reasonably accurate approximation for the actual elastic modulus that needs to be used (Bessmertnykh and Dontsov, 2018). Also note that the PKN model assumes that the fracture is long, i.e. height is much smaller than length. This is often the case, but not always, e.g. at the Hydraulic Fracture Test Site 2 (HFTS-2) where fracture heights were observed to be greater than 1800 ft (Zoback et al., 2022). It is also possible to calculate the stress change in the formation, caused by the single fracture (Sneddon, I.N., 1946):

$$\begin{aligned} \Delta S_h &= P_n \left[1 - \frac{x^3}{(x^2 + (H/2)^2)^{3/2}} \right], \\ \Delta S_H &= \nu (\Delta S_h + \Delta S_v), \\ \Delta S_v &= P_n \left[1 - \frac{x^3 + 2x(H/2)^2}{(x^2 + (H/2)^2)^{3/2}} \right]. \end{aligned}$$

The above quantities represent the change of stress along the x axis, which is the line perpendicular to the fracture plane that passes through the center of the vertical fracture, as indicated in Fig. 5. The right side of

the same figure plots the change of these principal stresses. As can be seen from the result, all stresses are affected by the fracture and the disturbance decreases at distances that are comparable with fracture height. The minimum principal stress change is always positive (compression), while the vertical stress features a positive difference that at some point becomes negative (or tensile). The maximum horizontal stress is changed solely due to Poisson's effect. The value of Poisson's ratio of 0.25 is used to plot the figure.

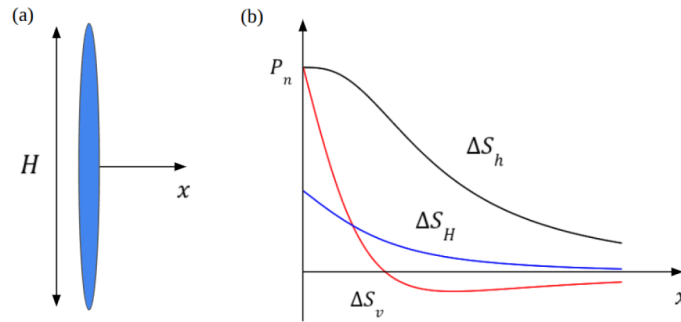


Figure 5—Change of stress from a single vertical hydraulic fracture with constant height H . (a): Schematic of a single vertical fracture. (b): Variation of principal stress components away from the fracture.

When multiple fractures are induced on a single horizontal well, this leads to the accumulation of stresses, see e.g. Roussel (2017). In the field, this typically translates into increasing ISIP values along the well. When fracture spacing is small compared to the fracture height (which is typically the case in field operations), then the total stress accumulation can be estimated as

$$\Delta S_h = \frac{2E\bar{V}}{HA} \int_0^\infty \left[1 - \frac{x^3}{(x^2 + (H/2)^2)^{3/2}} \right] dx = \frac{2E\bar{V}}{\pi A},$$

$$\Delta S_H = \nu \Delta S_h$$

$$\Delta S_v = \frac{2E\bar{V}}{HA} \int_0^\infty \left[1 - \frac{x^3 + 2x(H/2)^2}{(x^2 + (H/2)^2)^{3/2}} \right] dx = 0.$$

To obtain the above equations, the fracture volume is represented as $V = \bar{V}dx$, where $\bar{V}$ is fluid loading (i.e. volume of fluid injected per unit length of the well), while dx is cluster spacing. Since the latter is considered small, it is used as a differential to calculate the integral. Also note that the integration is to infinity, which essentially means that the well length is much greater than the fracture height. The above result demonstrates that the minimum horizontal stress change is directly proportional to fluid loading, Young's modulus, and inversely proportional to fracture area. Interestingly, there is no total accumulation of vertical stress, while the maximum horizontal stress change is a fraction of ΔS_h . While the absence of vertical stress accumulation is a mathematical fact for given model assumptions, it can also be explained by noting that vertical hydraulic fractures cause predominantly uniaxial compression of the reservoir in the direction of S_h . Further, given the assumption of relative smallness of fracture height relative to length, the deformation pattern is effectively two-dimensional (as viewed from the top) and in the state of plane stress (since height is small). The latter plane stress state implies constant stress in the vertical direction, which is equivalent to zero stress change. Note that this result is an estimate that is based on the assumption of identical constant height fractures placed uniformly on a horizontal well. The stress shadow interaction between the fractures likely leads to more complex shapes and reduced value of the total stress growth. While the numeric multiplier can be reduced, the scaling relationship with respect to parameters is expected to hold.

For instance, for 50 bbl/ft fluid loading, 6 Mpsi rock, and fracture with the total length of $L = 2000$ ft and height of $H = 500$ ft, the total stress build up is

$$\begin{aligned}\Delta S_h &= \frac{2E'\bar{V}}{\pi A} \approx 1000 \text{ psi}, \\ \Delta S_H &= \nu \Delta S_h \approx 250 \text{ psi}, \\ \Delta S_v &= 0 \text{ psi}.\end{aligned}$$

Therefore, for a single well with such parameters, the expected stress accumulation is up to 1000 psi in the S_h direction and 0 psi in the S_v direction. This demonstrates the ability of operations, even with a single horizontal well, to substantially change the state of stress, particularly in the minimum stress direction. This can bring the total minimum horizontal stress S_h closer to the vertical one, which in turn can lead to the initiation of horizontal fractures. Note again that these are very rough estimates that are intended to demonstrate the potential order of magnitude for the stress change.

For pads, the situation is getting worse in the sense that higher stress accumulations are possible. Let's consider a pad consisting of N wells in the horizontal direction and M wells in the vertical direction, and let's further assume that all the wells are fractured at approximately the same time in a zipping sequence. In this situation, the total fluid loading is increased because there are multiple wells, but the total effective area is also increased. Note that we need to look at the 'collapsed' area, or area that is obtained by looking at the fractures from the gun barrel view. For the case of $l = 800$ ft horizontal well spacing and $h = 300$ ft vertical well spacing, we obtain

$$\begin{aligned}\Delta S_h &= \frac{2E'\bar{V}NM}{\pi(L+(N-1)l)(H+h(M-1))} \approx 4500 \text{ psi}, \\ \Delta S_H &= \nu \Delta S_h \approx 1120 \text{ psi}, \\ \Delta S_v &= 0 \text{ psi}.\end{aligned}$$

Note that the final numbers are obtained assuming $N \gg 1$ and $M \gg 1$, which represents the worst case scenario. For the latter case, the total stress accumulation is

$$\Delta S_h = \frac{2E'\bar{V}}{\pi lh}.$$

The difference with the corresponding equation for a single well is that the fracture area is replaced by the area defined by well spacings. For the single well case, higher fluid loadings also lead to more surface area, in which case there is no drastic increase in the stress accumulation. For the pad case, on the other hand, there is nowhere for fluid to go and therefore the stress accumulation can substantially increase, as is demonstrated in this example. It is also interesting to observe that $\bar{V}/(hl)$ is the volumetric strain or the volume of fluid injected per volume of rock. Therefore, the above equation states that the stress change is proportional to the elastic modulus and strain induced by injection.

To evaluate the accuracy of the above estimates, Fig. 6 presents results from numerical simulation for a single well. Simulations were performed using a coupled hydraulic fracturing and reservoir simulator (McClure et al., 2025). Fluid loading was approximately equal to $\bar{V} = 45$ bbl/ft, while Young's modulus was $E' = 3.4$ Mpsi, and Poisson's ratio was $\nu = 0.25$. Given the fracture geometry, this translates to $\Delta S_h = 280$ psi, $\Delta S_H = 70$ psi, and $\Delta S_v = 0$ psi estimated stress shadow. The top left picture in the figure shows the fracture induced stress ΔS_h , the top right picture shows ΔS_H , while the bottom picture illustrates the variation of ΔS_v around fractures. As can be seen from the results, the estimate for both of the horizontal stress components is relatively accurate, especially given the variability of fracture geometry. The vertical stress change is consistent with the result shown in Fig. 5, i.e. it changes sign away from the fractures. The magnitude of ΔS_v is also an order of magnitude smaller than that for ΔS_h , which is also consistent with the estimate. Note that the maximum fracture extension was used for the value of the total length. Using a smaller length would

increase the value of ΔS_h . Also note that in this example, the fracture height was relatively constant, which justifies the constant height assumption used to derive the estimate.

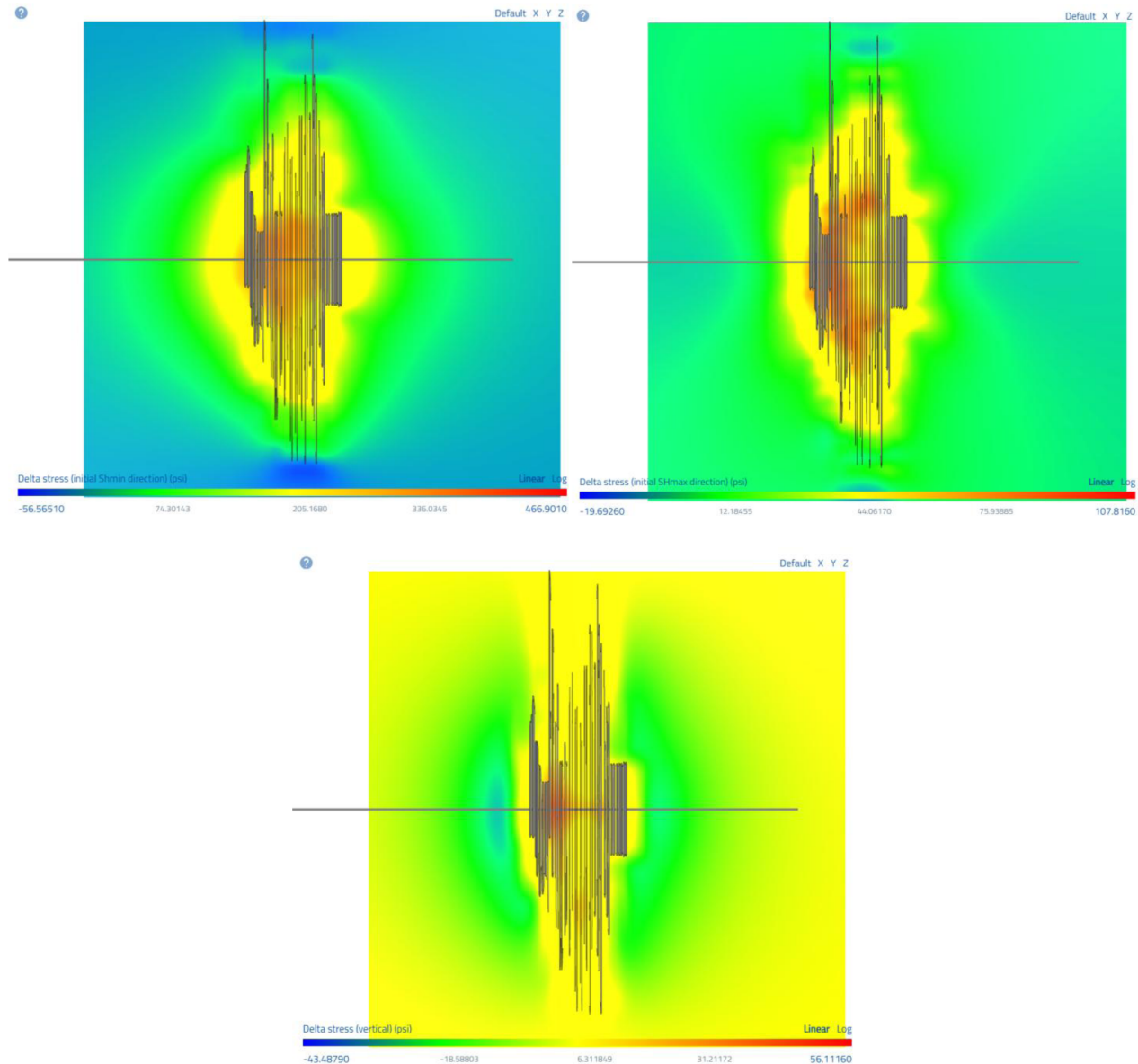


Figure 6—Numerical simulation of multiple hydraulic fractures on a single well and the associated induced stress. Top left: distribution of fracture induced horizontal stress ΔS_h in a horizontal plane. Top right: distribution of fracture induced vertical stress ΔS_H in a horizontal plane. Bottom: distribution of fracture induced vertical stress ΔS_v in a horizontal plane.

Further, Fig. 7 presents results from numerical simulation for a pad consisting of nine wells. The same fluid loading per well was used, i.e. $\bar{V} = 45$ bbl/ft, and rock properties were also the same, i.e. Young's modulus was $E' = 3.4$ Mpsi, and Poisson's ratio was $\nu = 0.25$. The resultant estimates for the stress shadow are $\Delta S_h = 1100$ psi, $\Delta S_H = 280$ psi, and $\Delta S_v = 0$ psi. These estimates are consistent with the values shown in the figure, thus demonstrating applicability of the estimate as well as an increase of the overall stress shadow magnitude caused by the closely spaced wells.

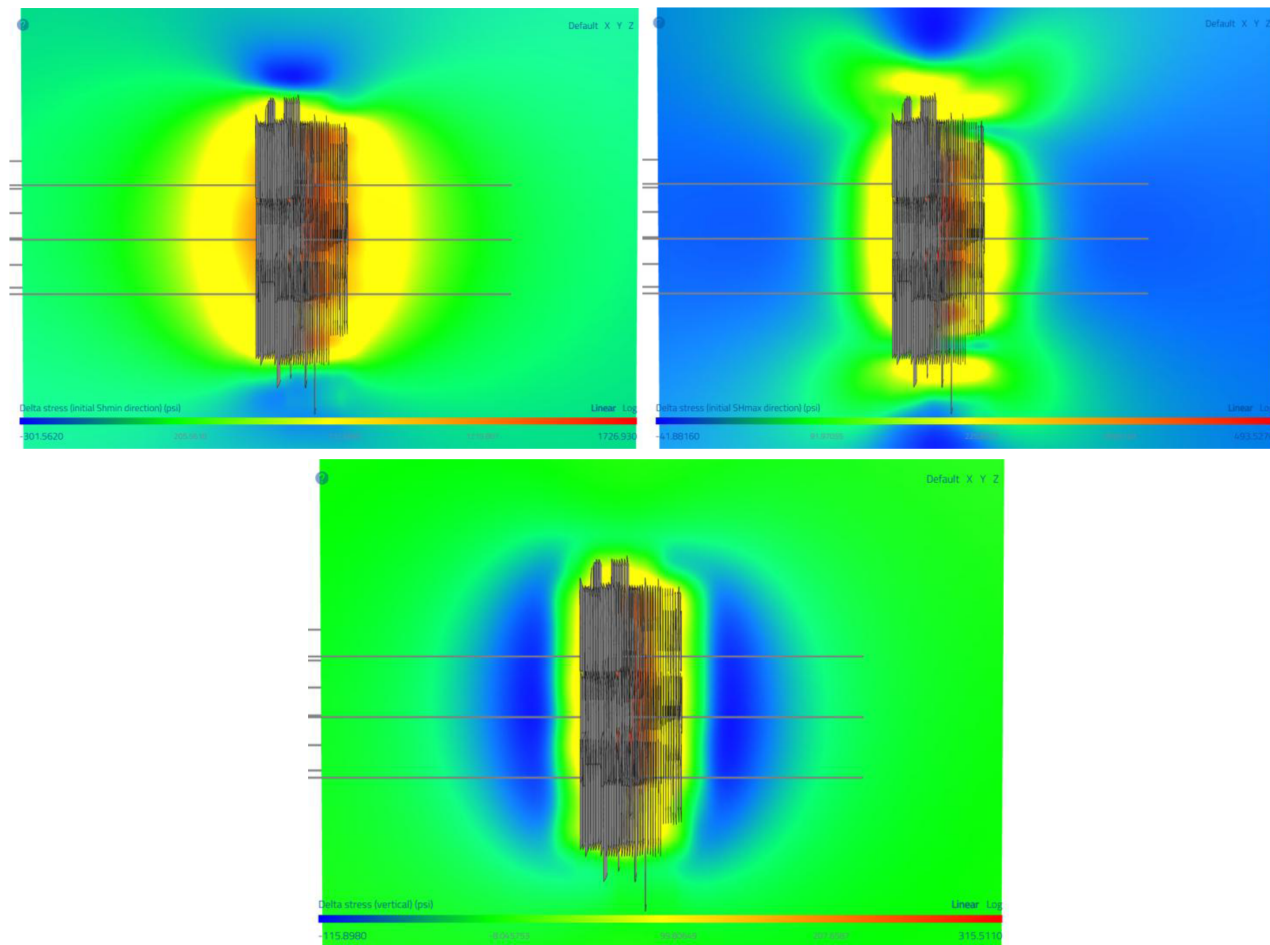


Figure 7—Numerical simulation of multiple hydraulic fractures within a pad and the associated induced stress. Top left: distribution of fracture induced horizontal stress ΔS_h in a horizontal plane. Top right: distribution of fracture induced vertical stress ΔS_v in a horizontal plane. Bottom: distribution of fracture induced vertical stress ΔS_v in a horizontal plane.

One of the assumptions for the above estimates was the fact that fluid stays inside the fracture. In reality, the rock is porous and fluid leaks into the formation. Let's consider another limiting scenario, in which all fluid quickly leaks into formation and fractures close. Under the assumption of smaller height than length, i.e. similar to PKN fracture model, the solution for a uniformly pressurized cuboid in isotropic poroelastic solid can be used to calculate the minimum horizontal stress increase as (Nowacki, 1986; Dontsov, 2021)

$$\Delta S_h = \alpha \frac{1-2\nu}{1-\nu} \frac{M \bar{V}}{A},$$

$$\Delta S_v = 0.$$

Here α is Biot coefficient, M is Biot modulus, while $\bar{V}$ is fluid loading and A is the cross-sectional area, which is the same as for the case with fractures. The latter area is equal to hl for the case of a pad with many wells. Fundamentally, there is not much of a difference between this expression and the previous one that considered fractures. Both are proportional to fluid loading per unit area or the volume of fluid injected per volume of rock. Both are also proportional to stiffness of the rock. The difference comes in the dimensionless multiplier, which can be noticeably smaller for the case of fluid leak-off. This difference can be important for practical applications, but what is interesting is that the same operational parameters affect the result.

This section presented estimates for the potential change of the state of stress caused by hydraulic fracturing operations. As was also pointed out before, here we provide estimates that are based on numerous assumptions, so the results should not be taken literally. Instead, the results demonstrate the potential of high stress alteration by operations, and the level of change increases for multi well pads relative to a single

well. The minimum horizontal stress change can reach 1000s of psi and therefore, given the initially small stress anisotropy, this can lead to the initiation of horizontal fractures at some point within the pad. This demonstrates that operations can also contribute to the formation of horizontal fractures, and this also means that horizontal fractures can be potentially controlled by operational parameters.

Discussion

Previous sections clearly demonstrate that the potential for having horizontal fractures depends on both rock properties and operational parameters. This is an interesting observation because it implies that simply translating the same operational procedures from one field to another may cause horizontal fractures, i.e. may lead to qualitatively different stimulations. At the same time, sensitivity to operational parameters opens the possibility to manage horizontal fractures by changing treatment design as needed.

As a starting point, let's consider a simple calculation that can be used to estimate if horizontal fractures will initiate or not. Consider a formation with the fracture gradient of 0.8 psi/ft and the vertical stress gradient of 1.1 psi/ft. Let the well depth be 7000 ft. In this case, the minimum horizontal stress is $S_h = 5600$ psi and the vertical stress is $S_v = 7700$ psi. For the tensile strength, let's take $T_h = 1000$ psi and $T_v = 0$ psi for simplicity. In this situation, since $S_h + T_h = 6600$ psi < 7700 psi $= S_v + T_v$, vertical hydraulic fractures will initiate at the toe of the well. As stress shadow builds up, the total minimum horizontal stress is increasing, while the vertical stress remains the same. For horizontal fracture initiation, we now need to compare $S_h + T_h + \Delta S_h$ and $S_v + T_v$, where ΔS_h is the horizontal stress build up. For the given example, if the total stress shadow accumulation exceeds 1100 psi, then horizontal fractures will form. To be clear, in this example fracture initiation is considered at a perforation cluster (in contrast to a T-shaped fracture, for which the horizontal fracture initiates from a vertical fracture). For the initiation at a weak interface, i.e. when a vertical hydraulic fracture reaches a bedding plane and initiates a horizontal fracture, we need to make sure that the fluid pressure in the fracture exceeds the total vertical stress and tensile strength, i.e. $P > S_v + T_v$. The latter fluid pressure is equal to the sum of the total minimum horizontal stress and the net pressure: $P = S_h + \Delta S_h + P_n$. This example demonstrates how both rock properties (tensile strength anisotropy) and operational procedures (fracture induced stress or stress shadow) can lead to horizontal hydraulic fractures even in situations, in which the difference between the minimum horizontal and vertical stresses is initially significant. It is also important to note that initiation of horizontal fractures does not automatically imply large-scale propagation of horizontal fractures since the latter is affected by the state of stress, as is demonstrated next.

Next, let's revisit Marcellus' example (Alalli and Zoback, 2018) that was mentioned in the introduction. One possible way to reproduce such results is to consider two rock layers with different minimum horizontal stress gradients and tensile strength anisotropy. In particular, consider the following parameters: vertical stress gradient 1.1 psi/ft, horizontal stress gradient in the layer above is 0.9 psi/ft, horizontal stress gradient in the layer below is 1.0 psi/ft, well depth is approximately 7000 ft (well is dipping and crosses both layers), tensile strength is $T_h = 1000$ psi, while $T_v = 0$ psi. For these numbers, the initiation pressure for vertical fractures is 7300 psi in the layer above and 8000 psi in the layer below. At the same time, the initiation pressure for horizontal fractures is approximately 7700 psi for both layers. This indicates that vertical hydraulic fractures should initiate in the layer above, while horizontal fractures should initiate in the layer below. Results of numerical simulation using parameters mentioned above are shown on the left in Fig. 8. As can be seen from the figure, horizontal fractures are indeed initiated at the lower portion of the well, but unable to propagate. The reason for this lies in the fact that it only takes additional 300 psi to initiate vertical fractures in the lower zone. Once the net pressure (and the accumulated stress shadow) of horizontal fractures exceeds this number, vertical fractures are initiated. But since the latter are exposed to lower confining stress, they tend to take most of the fluid and horizontal fracture propagation ceases. Once the horizontal stress is increased by 0.05 psi/ft, the situation changes (right picture in Fig. 8). Now only horizontal fractures form in the lower part of the well, but vertical fractures form in the upper layer. Note that

pure horizontal fractures are initiated despite $S_v > S_h$. This example indicates that under some circumstances horizontal fractures can initiate, but not propagate far. Therefore, it is not sufficient to simply look at the initiation condition. Stress contrast needs to be sufficient to prevent initiation of vertical fractures as the latter can win the competition for fluid distribution and arrest horizontal fracture growth.

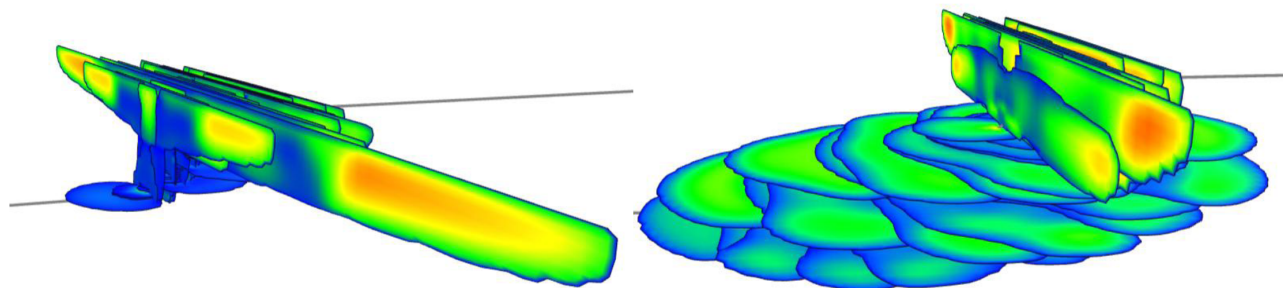


Figure 8—Numerical simulation for an inclined well that is drilled through two different formations. Color shows fracture aperture. The left side shows an example, in which horizontal fractures initiate, but do not propagate far. The right side shows the same simulation, but with the horizontal stress gradient increased by 0.05 psi/ft. For the latter case, horizontal fractures are dominant in the lower formation, while vertical fractures are dominant in the upper layer, which is consistent with observations in [Alalli and Zoback \(2018\)](#).

To proceed, consider a case that resembles the Eagle Ford example with a propped horizontal fracture that was discussed in the introduction. First of all, note that the initial state of stress corresponds to a normal faulting regime, i.e. $S_h < S_H < S_v$. The stress build up caused by vertical fractures increases both S_h and S_H . Thus, this can bring the state of stress to strike-slip regime, i.e. $S_h < S_v < S_H$. Within this strike-slip stress state, there is a competition between transverse vertical and horizontal fractures, while the longitudinal fractures are less favorable for such stress state. Moreover, in order to have both large scale vertical and horizontal hydraulic fractures simultaneously, we need to have $S_h \approx S_v$. Otherwise, if the stress contrast is significant, vertical hydraulic fractures will dominate as soon as they initiate, as is demonstrated in the previous example. Results of numerical simulations are shown in [Fig. 9](#) for the case in which the contrast between the vertical and minimum horizontal stress is only 0.01 psi/ft. Two scenarios are considered, one with some height growth (left) and another with strongly height-contained fractures (right). Two large horizontal fractures propagate in the former scenario, while only a single horizontal fracture is formed in the latter case. The color shows proppant mass per area. In both cases, proppant is able to travel a large distance away from the wellbore, which is consistent with the field observation. Note that the hydraulic length is almost double the propped length.

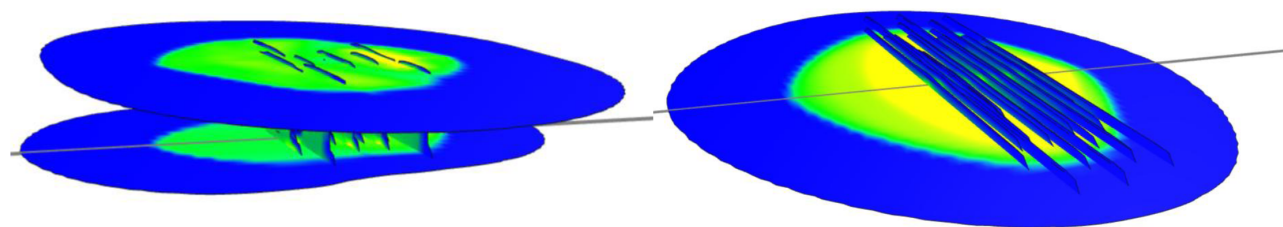


Figure 9—Numerical simulation, in which both vertical and horizontal fractures propagate simultaneously. Color shows proppant mass per area. Left: weaker stress barriers and more vertical growth. Right: stronger stress barriers and less vertical growth.

The last example illustrates how sub-horizontal hydraulic fractures can cause casing deformation. To model such a scenario, a sub-horizontal bedding interface is explicitly included. Results of numerical simulations are shown in [Fig. 10](#). Vertical hydraulic fractures propagate and reach the bedding plane. The fractures are being arrested by the interface, but given its finite size in the simulation, some fractures are

able to grow around it at the edges. Two scenarios are considered, in which the difference of the minimum horizontal stress gradient is only 0.02 psi/ft. For the case with lower S_h , the bedding plane gets pressurized due to fluid diffusion, but fluid pressure is insufficient to fully open it. In this case, sliding along the sub-horizontal plane is localized. In the second case with higher S_h , fluid pressure is sufficient to open the bedding plane. This results in sub-horizontal hydraulic fracture propagation. The zone of sliding is also much larger and reaches the vertical part of the well in this example, which can cause casing deformation. This example illustrates that it takes only a few 100s of psi of pressure to make the difference in the level of casing deformation. This can be explained by the fact that the net pressure of a horizontal fracture is only a few 100s psi. As a result, changing the total fluid pressure by 100 psi can change the net pressure from 0 (closed fracture) to 100 psi (open fracture). Also notice that fluid in the sub-horizontal fracture prefers to flow upwards along the dip direction. This is caused by the normal stress gradient, i.e. the normal stress in the preexisting fracture decreases with shallower depths. Consequently, if the bedding plane dips upwards towards the vertical section of the well, this can potentially cause more casing deformation.

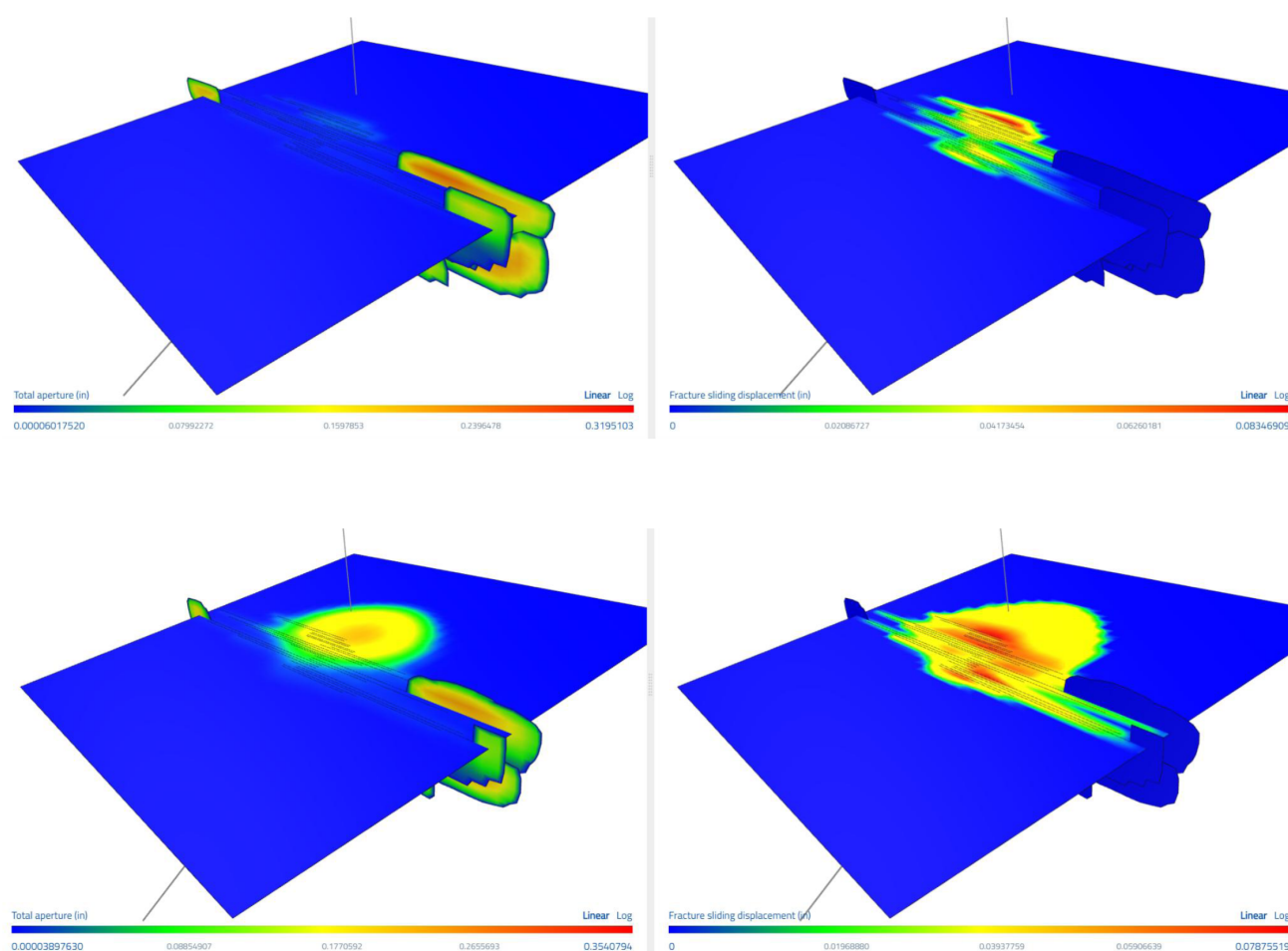


Figure 10—Results of numerical simulations for the case of sub-horizontal fracture sliding. Vertical hydraulic fractures induced from a horizontal well reach a sub-horizontal fracture and stimulate it. Left: total fracture aperture. Right: fracture sliding displacement. The model at the bottom has an increased minimum horizontal stress gradient by 0.02 psi/ft relative to the result at the top.

It is worth discussing potential mitigation strategies. Given the analysis presented in this paper, there is one operational parameter that affects the result, which is the volume of fluid injected per volume of stimulated rock. It can be changed by either changing the fluid loading and/or well spacing for pad stimulations. Also, the rock itself has a "maximum capacity" before fractures turn horizontal. The first step is therefore to find this "maximum capacity" by better understanding the state of stress and other

rock parameters, such as tensile strength and its anisotropy. The next step is to design hydraulic fracturing treatments that do not exceed the limit. This can be achieved by changing fluid loading or well spacing. Or, if horizontal fractures are of concern only near the vertical section of the well, then reduce fluid loading towards the heel of the well. Another possibility is to avoid simultaneous stimulation of many closely spaced wells within the same pad. One possibility is to stimulate a few wells, produce them for some time, and then stimulate some more wells nearby. Production will reduce the total fluid loading (and the associated stress change) experienced by the reservoir and can help to avoid horizontal fractures. Reduction of fluid loading and alternation of well stimulation sequence clearly affects production and economics. It is possible that the most economical case corresponds to some level of horizontal hydraulic fractures in the reservoirs, and therefore completely avoiding horizontal fractures may not be the ultimate goal.

Conclusions

The purpose of this paper is to explore physical mechanisms that can explain the field observations of horizontal hydraulic fracture propagation. While hydraulic fractures are often assumed to be vertical, there is also field evidence that demonstrates the existence of horizontal hydraulic fractures in some situations. In particular, horizontal hydraulic fractures were observed in Marcellus using microseismic (Alalli and Zoback, 2018), a propped horizontal hydraulic fracture was observed in a vertical core in Eagle Ford despite microseismic showed substantial height growth, simultaneous propagation of a sub-horizontal fracture and several vertical fractures was observed in Sichuan Basin using fiber optic measurements (Wang et al., 2025), and there are numerous casing deformation results that indicate sub-horizontal fracture sliding in the Vaca Muerta, Sichuan, Montney, Duvernay, Anadarko, Midland, and other locations (Uribe-Patino et al., 2024). Therefore, we can conclude that horizontal hydraulic fractures can occur in unconventional rock reservoirs and often simultaneously with vertical fractures. They can open, transport proppant, and ‘carry’ shear deformation or sliding for large distances.

Analysis in this paper indicates that the presence of horizontal fractures is affected by both rock properties and operational parameters. Shales are finely layered and anisotropic. Elastic anisotropy reduces net pressure of horizontal fractures and makes it easier to open the fractures. Fracture toughness for the fracture propagating along the weak bedding interface is also expected to be small. This, together with the fact that there are no natural barriers for horizontal fractures, leads to potentially large horizontal fracture dimensions. In addition, tensile strength is anisotropic in rocks. As a result, since the initiation criterion requires fluid pressure to exceed the total stress plus tensile strength in a given direction, it is possible to initiate a horizontal fracture even in situations when the vertical stress still exceeds the minimum horizontal stress. Thus, tensile strength anisotropy promotes initiation of horizontal fractures, while elastic anisotropy and toughness anisotropy extends fracture propagation. Finally, rocks exhibit viscoelastic properties, whereby the difference between the vertical and horizontal stresses can be greatly reduced for certain rock types. The next contributing factor is fluid injection into the subsurface, which changes the state of stress. The primary parameter that affects the level of change is the volume of fluid injected per stimulated reservoir volume. This is controlled by fluid loading per well as well as the well spacing for the case of a pad treatment. Estimates were made showing that high fluid loadings and closely spaced wells can lead to the stress build up of up to several 1000s of psi. This is sufficient to alter the stress state to an extent that horizontal fractures start forming.

Several examples are shown demonstrating that it is possible to qualitatively match field observations with horizontal fractures. One interesting observation is that the "window of opportunity" for horizontal fractures is relatively narrow. For instance, for the considered casing deformation example, the difference between significant sliding on the sub-horizontal interface and practically no sliding was only several 100s of psi of the minimum horizontal stress. This is explained by the fact that the net pressure of a single horizontal fracture is also on the order of a few 100s of psi. As a result, changing the total horizontal

stress and fracturing pressure by such a small amount is capable of opening a horizontal fracture. The open horizontal fracture propagates and, since it is unable to sustain shear stress, causes sliding at distances far away from the injection point. Finally, potential mitigation strategies include a better understanding of rock properties, and managing fluid loading. Given the state of stress and anisotropy, formations have a certain fluid loading capacity, beyond which horizontal fractures will form. Mathematical modeling can be used to predict operational constraints (such as fluid loading, well spacing, stimulation/production order) that determine whether horizontal fracture propagation and bedding plane slip will occur.

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